Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/10/2025
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of P21/CDKN1a Pathway in Microgravity-Induced Bone Tissue Regenerative Arrest - A Spaceflight Study of Transgenic P21/CDKN1a Null Mice in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
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Key Personnel Changes/Previous PI:	NA		
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Task Description:	Tissue degeneration and the failure of normal tissue regenerative processes in microgravity are increasing concerns for sustaining life in space as the duration of future flight missions increases. Specifically, our laboratory has identified normal adult stem cell-based tissue regenerative processes, such as the formation of new bone, cartilage, immune cells, and blood from bone marrow hematopoietic stem cell precursors, as being particularly sensitive to the stresses of spaceflight in microgravity. The cellular mechanisms activated in mouse bone during spaceflight are not completely understood, but they include induction of oxidative stress and the inhibition of differentiation of marrow mesenchymal stem cells, via activation of CDKN1a/p21-mediated cell cycle arrest in proliferative osteoprecursor cells on the bone surface. The identification of CDKN1a/p21 mRNA and protein as being highly up-regulated by spaceflight in mice is a seminal finding, because this gene is a master regulator of tissue regeneration, that when knocked out allows for mammalian limb regeneration similar to that found in amphibians. Because we established in multiple spaceflight and ground experiments that the CDKN1a/p21 pathway is a key molecular mechanism of inhibiting tissue regenerative processes specifically activated in microgravity, we sought to fly a CDKN1a/p21 null mice in microgravity to test the hypothesis that this gene is required for the arrest of regenerative bone formation observed in microgravity. Specifically, we hypothesized that CDKN1a/p21 is the key effector of cell cycle arrest in bone tissue response to the stresses of spaceflight in microgravity and more broadly that CDKN1a/p21 is required for bone regenerative arrest in mice during spaceflight. To test this hypothesis, we proposed the following specific goals: 1) To determine the extent of osteoblasts proliferation, differentiation, and mineral deposition in microgravity compared to 1g in a CDKN1a/p21-null background versus WT 2) To determine if the coupling and balance of increased osteoclastogenesis and decreased osteoblastogenesis in microgravity is altered in a CDKN1a/p21-null background 3) To determine if CDKN1a/p21 expression in microgravity is regulated via mRNA stabilization by the microgravity-induced Cold Induced RNA Binding Protein (CIRBP) in response to oxidative stress in bone. The experimental approaches originally proposed in 2014 to achieve these aims include MicroComputed tomography, bone histology, and histomorphometry of injected Calcein plus BromodeoxyUridine labels to determine bone growth, and bulk bone tissue genomics. We have however also recently updated our analysis plans to include novel methodologies that now allow us to quantify at the single cell level gene expression in the bone marrow stem cell niche containing various stages of differentiation of the osteoprogenitor and hematopoietic lineages. The combined results of these studies will allow us to definitively determine if the presence of CDKN1a/p21 during spaceflight induces arrest of bone growth and regeneration mediated by CDKN1a/p21 inhibition of osteoblastic proliferation and differentiation, and if so, how this might be coupled to osteoclastic activity induced in space. In addition, for the first time, these studies seek to connect the potential effector molecular mechanism of regenerative arrest in bone (by CDKN1a/p21) to its inducer, hypothesized here to be oxidative (and other) stress response gene CIRBP stabilization of CDKN1a/p21 message RNA. When completed, these studies will provide molecular mechanistic evidence for how CDKN1a/p21, one of the most important master regulatory genes of stem cell regenerative health, is modulated by microgravity and space radiation, and could be the basis for development of preventive treatments designed to counter tissue regenerative deficits in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Recent advances in the scientific understanding of stem cells and their role in tissue regeneration created a new field of regenerative medicine that holds great promise both in the treatment, as well as prevention, of many degenerative diseases often associated with aging. In short duration human spaceflight, when the body is weightless and unloaded in microgravity, multiple tissue degenerative conditions occur, such as muscle atrophy, bone loss, immune deficiencies, and others that may have a deficit in stem cell-based tissue regeneration in common. Likewise, on Earth, when we increase mechanical loading of the body with exercise we observe an increase in adult stem cell-based tissue regeneration, such as in bone strengthening, muscle mass increase, and increased production of immune cells and red blood cells from hemotopoeietic stem cell progenitors. In total, spaceflight and ground experimental evidence from human and animal studies now strongly suggests that stem cell-based regenerative health is regulated by mechanical loading of the body, increasing with exercise and load, and decreasing with inactivity and unloading, as in microgravity. Our proposed experimental work in microgravity flight experiments and mechanical unloading models addresses the molecular, cellular, and tissue aspects of the hypothesis that gravity mechanical loading regulates regenerative health. The novel results we obtained identify the transition between stem cell progenitors and differentiated cells as a likely cellular process regulated by mechanical loading and specific molecular candidates such as the p21/CDKN1a pathway induced in microgravity as inhibitory of stem cell-based tissue regeneration in bone. This work is highly significant to NASA space exploration and medicine on Earth because it holds the promise of greater scientific understanding of how to maintain and monitor health during future voyages of space exploration, and solidly establishes Earth gravity as a major shaper of human life through promoting tissue regenerative health.
Task Progress:	Despite staffing limitations, regarding technical and scientific progress of the grant, we can report that the grant has successfully accomplished all its Scientific Aims, including making the novel and important finding that p21/Cdkn1a is mechanistically involved in negatively regulating both osteogenesis and osteclastogenesis in space (Aims 1 and 2). Our data include multiple solid lines of evidence that p21/Cdkn1a acts as moderator of the induction of osteogenesis, and also as moderator/inhibitor of the induction of osteoclast lineage differentiation, both in mechanical loading on Earth and weightlessness unloading in space. Our experiments both with isolated primary osteogenic cultures under stretch mechanical stress and mice in space, plus on Earth in hindlimb unloading and running wheel exercise, show that, in the absence of p21/Cdkn1a, the knockout mice have highly increased osteogenic Calcein deposition responses and coupled also highly increase osteoclastogenesis endosteal pitting responses. These findings are further supported by our experiments with single cell genomic analysis of the bone marrow mesenchymal and hematopoietic stem cell progenitors, which also shows p21 elevation in space both in stromal osteoprogenitors as well as the monocytic osteoclast progenitor lineage. Finally, x-ray high-resolution microcomputed tomography imaging of the load bearing femur shaft region, from where single cell samples were collected, show not only bone loss in wild-type (WT) mice in space, but even more dramatic losses in p21/Cdkn1a null mice also space flown. Beyond these analyses in the fall of 2024, we also completed histology sectioning of femur samples and are currently conducting confocal imaging quantitative analyses of Red Blood Cells in the femur marrow compartment, plus Tartrate-Resistant Acid Phosphatase (TRAP) staining for osteoclasts in the endosteal surface and marrow proliferation. Regarding scientific productivity, the Rodent Research-10 (RR-10) grant has already resulted in several peer reviewed manuscripts being published; others are in progress. [Ed. Note: Please see Cumulative Bibliography.]
Bibliography Type:	Description: (Last Updated: 12/08/2025)
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Abstracts for Journals and Proceedings	Eduardo A.C. Almeida(Ames Research Center Mountain View, California, United States) Rukmani Cahill(Rensselaer Polytechnic Institute Troy, New York, United States) Yasaman Shirazi(Ames Research Center Mountain View, California, United States) Shiyin Lim(University of California, Berkeley Berkeley, California, United States) Cassandra Marie Juran(Blue Marble Space Institute of Science Seattle, Washington, United States) Elizabeth A Blaber(Rensselaer Polytechnic Institute Troy, New York, United States) "Deletion of Cdkn1a in Mice Rescues Reduced Osteogenesis in Microgravity and Increases Coupled Osteoclastic Bone Degenerative Activity and Bone Loss" Meeting: Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) Location: Washington, DC Country: US https://ntrs.nasa.gov/citations/20230010134 , Nov-2023
Abstracts for Journals and Proceedings	Shiyin Lim(University of California, Berkeley Berkeley, California, United States) Joanna E Veres(University of California, Berkeley Berkeley, California, United States) Eduardo Almeida(Ames Research Center Mountain View, California, United States) Grace D O'Connell(University of California, Berkeley Berkeley, California, United States) "Murine Spine Composition and Mechanics are Maintained After 30-Days of Spaceflight" Meeting: ASGSR Conference 2023 Location: Washington DC Country: US https://ntrs.nasa.gov/citations/20230010143 , Nov-2023
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